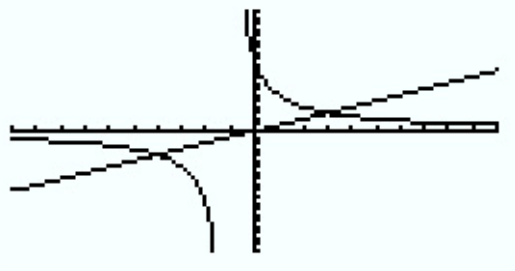


1.

Graphical Solution

From the GC

$$\therefore -4 \leq x < -1 \text{ or } x \geq 3$$

Algebraic Solution

$$\frac{12}{x+1} \leq x \Rightarrow \frac{12 - (x)(x+1)}{(x+1)} \leq 0$$

$$\Rightarrow \frac{12 - (x^2 + x)}{x+1} \leq 0$$

$$\Rightarrow \frac{x^2 + x - 12}{x+1} \geq 0$$

$$\Rightarrow (x+4)(x-3)(x+1) \geq 0$$

$$\therefore -4 \leq x < -1 \text{ or } x \geq 3$$

$$\frac{12}{|x|+1} \leq x$$

Note that the above inequality cannot hold for all $x < 0$

$$\frac{12}{|x|+1} \leq x \Rightarrow \frac{12}{x+1} \leq x \text{ as } x \geq 0.$$

$$\therefore x \geq 3$$

2(i).

$$(x-8)^{\frac{1}{3}};$$

$$= (-8)^{\frac{1}{3}} \left(1 - \frac{x}{8}\right)^{\frac{1}{3}}$$

$$= -2 \left(1 + \frac{1}{3} \left(-\frac{x}{8}\right) + \frac{\frac{1}{3} \left(-\frac{2}{3}\right)}{2!} \left(-\frac{x}{8}\right)^2 + \dots \right) = -2 \left(1 - \frac{1}{24}x - \frac{1}{576}x^2 + \dots \right)$$

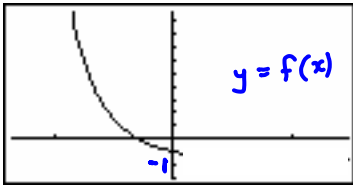
$$\text{Expansion is valid for } \left|\frac{x}{8}\right| < 1 \Rightarrow |x| < 8$$

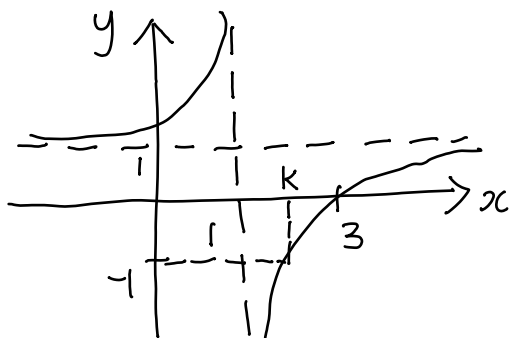
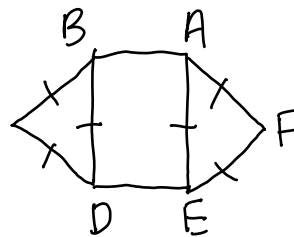
(ii)

$$\text{Let } x = -\frac{17}{8},$$

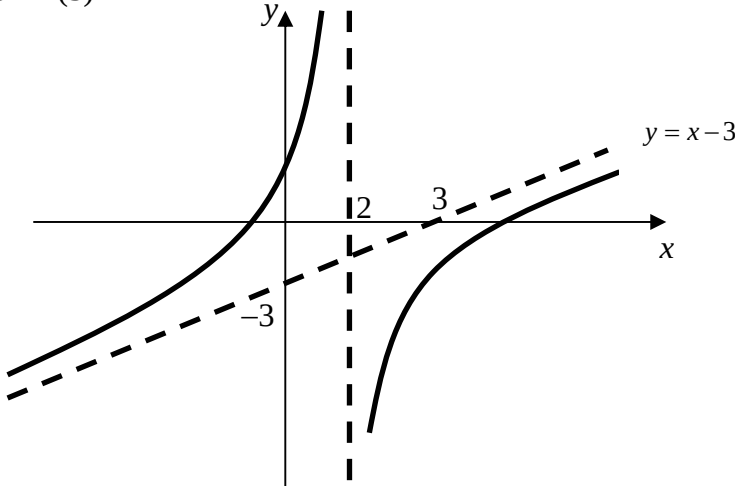
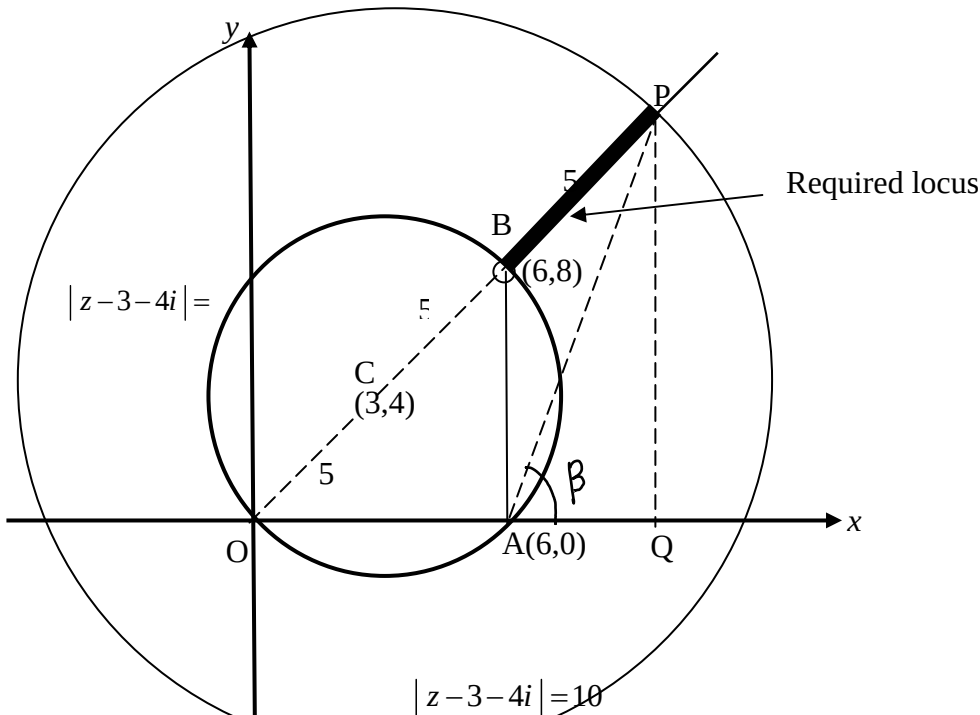
$$\left(-\frac{17}{8} - 8\right)^{\frac{1}{3}} = \left(-\frac{81}{8}\right)^{\frac{1}{3}} = -\frac{3}{2}\sqrt[3]{3} \approx -2(1 + 0.08854 - 0.00784)$$

$$\sqrt[3]{3} \approx 1.4409 \approx 1.441 \text{ (to 3 dp)}$$

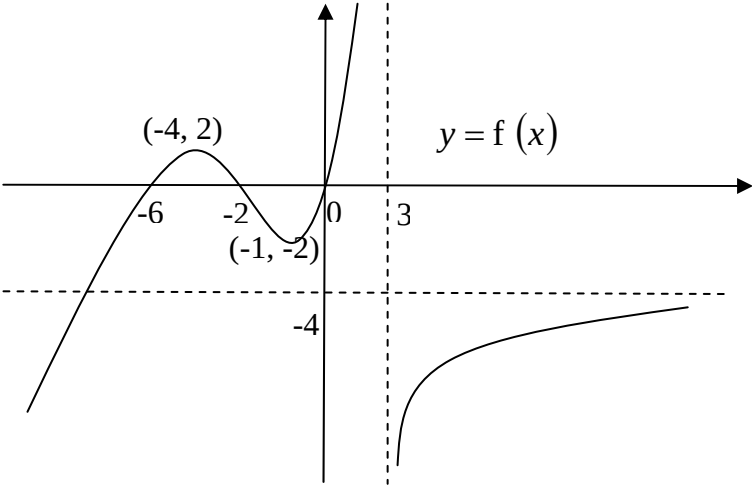
	H2 Maths Prelim Paper 1 Solutions	
3.	Let x, y and z be the number of packs of chicken, mutton and beef that Mrs Tan have to buy. $7.45x + 4.60y + 8.95z = 89.70 \quad \dots(1)$ $7.20x + 4.40y + 8.70z = 86.80 \quad \dots(2)$ $7.15x + 4.45y + 8.90z = 87.40 \quad \dots(3)$ Using GC, the RREF Matrix is $\begin{pmatrix} 1 & 0 & 0 & 6 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 4 \end{pmatrix}$ Mrs Tan has to buy 12 kg of chicken, 1 kg of mutton and 4 kg of beef.	
4.	Given : $T_1 = a > 0$ $T_2 = a(a-1) > 0$ For both T_1 and T_2 to be true, $a > 1$ ----- (1) Common ratio $r = \frac{T_2}{T_1} = a-1$ For the series to converge, $r < 1$ $\Rightarrow -1 < a-1 < 1$ $\Rightarrow 0 < a < 2$ Considering eqn (1): $\therefore 1 < a < 2$ Given : $T_n = T_{n+1} + T_{n+2} + \dots$ to ∞ $\Rightarrow ar^{n-1} = ar^n + ar^{n+1} + \dots$ $\Rightarrow \frac{ar^n}{r} = \frac{ar^n}{1-r}$ $\Rightarrow r = 1-r$ $\Rightarrow r = \frac{1}{2}$ i.e. $a = \frac{3}{2}$ Thus $S_\infty = \frac{a}{1-r} = \frac{\frac{3}{2}}{1-\frac{1}{2}} = 3$	
5(i).	$y = e^{x^2-2x} - 2$ $x^2 - 2x = \ln(y+2)$ $(x-1)^2 - 1 = \ln(y+2)$ $(x-1)^2 = 1 + \ln(y+2)$ $x = 1 \pm \sqrt{1 + \ln(y+2)}$ Since $x < 0$, $x = 1 - \sqrt{1 + \ln(y+2)}$ $D_{f^{-1}} = R_f = (-1, \infty)$ $\therefore f^{-1} : x \rightarrow 1 - \sqrt{1 + \ln(x+2)}, x > -1$ 	

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(iii)	$R_g \subseteq D_{f^{-1}} \Rightarrow R_g \subseteq (-1, \infty)$  $(2, \infty) \xrightarrow{g} (-1, 1) \xrightarrow{f^{-1}} (1 - \sqrt{1 + \ln 3}, 0)$	When $g(x) = -1$ $g(2) = \frac{2-3}{2-1} = -1$ $D_g = (2, \infty)$ Least $k = 2$	
6.	$AB = DE = \frac{r-4s}{2}$ Area $A = (s)(s) \sin(60^\circ) + \left(\frac{r-4s}{2}\right)s$ $= \left(\frac{\sqrt{3}}{2} - 2\right)s^2 + \frac{rs}{2}$ $\frac{dA}{ds} = \sqrt{3}s + \frac{r}{2} - 4s$ For maximum area, $\frac{dA}{ds} = \sqrt{3}s + \frac{r}{2} - 4s = 0 \Rightarrow s = \frac{r}{8-2\sqrt{3}}$ $\frac{d^2A}{ds^2} = \sqrt{3} - 4 < 0 \Rightarrow$ Area is at a maximum Maximum Area $= \left(\frac{\sqrt{3}}{2} - 2\right)\left(\frac{r}{8-2\sqrt{3}}\right)^2 + \left(\frac{r}{2}\right)\left(\frac{r}{8-2\sqrt{3}}\right)$ $= \frac{r^2}{2(8-2\sqrt{3})} \left(\frac{\sqrt{3}-4}{8-2\sqrt{3}} + 1\right)$ $= \frac{r^2}{4(8-2\sqrt{3})} = \frac{r^2(4+\sqrt{3})}{104}$ 		
7(a).	$\frac{d}{dx} \left[\cos^{-1} \left(\frac{1}{x^2} \right) \right] = \frac{-1}{\sqrt{1 - \left(\frac{1}{x^2} \right)^2}} \left(-\frac{2}{x^3} \right)$ $= \frac{2}{x\sqrt{x^4 - 1}}$		
7(b).	$x = \ln(\sin t) \Rightarrow \frac{dx}{dt} = \frac{\cos t}{\sin t}$		

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	$y = \cot t \Rightarrow \frac{dy}{dt} = -\operatorname{cosec}^2 t$ $\text{Therefore } \frac{dy}{dx} = -\frac{1}{\sin^2 t} \times \frac{\sin t}{\cos t} = -\frac{2}{2 \sin t \cos t} = -\frac{2}{\sin 2t} \quad (\text{proved})$	
	$\text{Gradient of normal} = \frac{1}{2} \sin 2t = \frac{1}{2}$ $\Rightarrow \sin 2t = 1$ $\Rightarrow 2t = \frac{\pi}{2} \Rightarrow t = \frac{\pi}{4}$ $\text{Therefore } x = \ln \left(\sin \frac{\pi}{4} \right) = \ln \left(\frac{\sqrt{2}}{2} \right) = -\frac{1}{2} \ln 2$ $\therefore y = 1$ $\text{Equation of normal: } y - 1 = \frac{1}{2} \left(x - \left(-\frac{1}{2} \ln 2 \right) \right) \Rightarrow y = \frac{1}{2} \left(x + \frac{1}{2} \ln(2) \right) + 1$	
	$PQ = RQ$ $\Rightarrow \text{normal at } R \text{ is the reflection of the above normal in the x-axis}$ $\Rightarrow \text{equation of normal at } R \text{ is } \Rightarrow y = -\left(\frac{1}{2} \left(x + \frac{1}{2} \ln(2) \right) + 1 \right)$ $\Rightarrow y = -\frac{1}{2} \left(x + \frac{1}{2} \ln(2) \right) - 1$	
8(a)	$\int e^{2x} \tan^{-1}(e^{-2x}) \, dx$ $= \frac{1}{2} e^{2x} \tan^{-1}(e^{-2x}) - \int \frac{1}{2} e^{2x} \frac{-2e^{-2x}}{1+(e^{-2x})^2} \, dx$ $= \frac{1}{2} e^{2x} \tan^{-1}(e^{-2x}) + \int \frac{1}{[1+e^{-4x}]} \, dx$ $= \frac{1}{2} e^{2x} \tan^{-1}(e^{-2x}) + \int \frac{e^{4x}}{e^{4x}+1} \, dx = \frac{1}{2} e^{2x} \tan^{-1}(e^{-2x}) + \frac{1}{4} \ln(e^{4x}+1) + C$	
8(b)	$\int \frac{x}{\sqrt{1-4x-2x^2}} \, dx = -\frac{1}{4} \int \frac{-4-4x}{\sqrt{1-4x-2x^2}} \, dx - \int \frac{1}{\sqrt{1-4x-2x^2}} \, dx$ $= -\frac{1}{2} \sqrt{1-4x-2x^2} - \int \frac{1}{\sqrt{2} \sqrt{\frac{3}{2} - (x+1)^2}} \, dx$ $= -\frac{1}{2} \sqrt{1-4x-2x^2} - \frac{1}{\sqrt{2}} \sin^{-1} \frac{\sqrt{2}(x+1)}{\sqrt{3}} + C$	
9(i).	$y = \frac{x^2 - ax + b}{x - 2} \Rightarrow x + (2 - a) + \frac{b + 4 - 2a}{x - 2}$ $\frac{dy}{dx} = 1 - \frac{b + 4 - 2a}{(x - 2)^2} = 0$ $(x - 2)^2 = b + 4 - 2a \Rightarrow x^2 - 4x + (-b + 2a) = 0$	

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	For C to have 2 stationary points, Discriminant: $(-4)^2 - 4(1)(2a - b) > 0$ $\Rightarrow b + 4 - 2a > 0 \Rightarrow b > 2a - 4$ (shown)	
(ii)	From (i), when $a = 5$, we know that the C has no stationary points since $b < 0 \Rightarrow b < 2(5) - 4$  $2x^2 - 10x - 1 = (mx - 3)(2x - 4) \Rightarrow \frac{x^2 - 5x - \frac{1}{2}}{x - 2} = mx - 3$ Therefore $a = 5, b = -\frac{1}{2}$. Plot $y = mx - 3$ on the same diagram. For equation $2x^2 - 10x - 1 = (mx - 3)(2x - 4)$ to have exactly 2 roots, $m < 1$. $\therefore$ The largest possible value of k is 1.	
10.	$5 \leq z - (3 + 4i) \leq 10$, $\arg(z - (6 + 8i)) = \arg(3 + 4i)$  Required locus	

	H2 Maths Prelim Paper 1 Solutions	
(i)	Max value of $z - 6 = AP$ By similar triangles, $\frac{OP}{OB} = \frac{PQ}{BA} \Rightarrow \frac{15}{10} = \frac{PQ}{8}$ $PQ = 12 \Rightarrow OQ = 9 \Rightarrow AQ = 3$ Max value of $z - 6 = AP = \sqrt{3^2 + 12^2} = \sqrt{153}$	
(ii)	Range of values of $\arg(z - 6)$ is $\beta \leq \arg(z - 6) < \frac{\pi}{2}$ $\beta = \tan^{-1}\left(\frac{12}{3}\right) = 1.33$ $\Rightarrow 1.33 \leq \arg(z - 6) < \frac{\pi}{2}$	
11a)	$y = \frac{1}{f\left(3 - \frac{x}{2}\right)}$ Points marked: $(8, -1/2)$, $(14, 1/2)$ Asymptotes: $y = -1/4$, $x = 6$, $x = 10$, $x = 18$	
(b)	$y^2 = -f\left(3 - \frac{x}{2}\right)$ Points marked: $(8, \sqrt{2})$, $(8, -\sqrt{2})$ Asymptotes: $y = 2$, $y = -2$, $x = 6$, $x = 10$, $x = 18$	

11 (c).		
12(i)	When $n = 1$, L.H.S. = $u_1 = 15$ R.H.S. = $4(2)^2 - 1 = 15 \quad \therefore$ the statement is true for $n = 1$ Assume that the statement is true for $n = k$ i.e. $u_k = 4(k+1)^2 - 1$ for some $k \in \mathbb{N}^+$ When $n = k + 1$, R.H.S. = $4(k+1+1)^2 - 1 = 4(k+2)^2 - 1$ L.H.S. = $u_{k+1} = u_k + 4 + 4\sqrt{u_k + 1}$ $ \begin{aligned} &= 4(k+1)^2 - 1 + 4 + 4\sqrt{4(k+1)^2 - 1 + 1} \\ &= 4(k+1)^2 - 1 + 4 + 4 \times 2(k+1) \\ &= 4[(k+1)^2 + 2(k+1) + 1] - 1 \\ &= 4[(k+1) + 1]^2 - 1 \\ &= 4(k+2)^2 - 1 \end{aligned} $ $\therefore$ The statement is true for $n = k + 1$ if it is true for $n = k$. Hence by the principle of induction, the statement is true for all $n \in \mathbb{N}^+$.	
(ii)	$ \begin{aligned} \frac{1}{u_n} &= \frac{1}{4(n+1)^2 - 1} \\ &= \frac{1}{[2(n+1)]^2 - 1} \\ &= \frac{1}{(2(n+1)-1)(2(n+1)+1)} \\ &= \frac{1}{(2n+1)(2n+3)} \end{aligned} $	

	$= \frac{1}{2(2n+1)} - \frac{1}{2(2n+3)}$ $\therefore A = \frac{1}{2}, B = -\frac{1}{2}$	
(iii)	$S_N = \sum_{n=1}^N \frac{1}{u_n}$ $= \sum_{n=1}^N \left(\frac{1}{2(2n+1)} - \frac{1}{2(2n+3)} \right)$ $= \frac{1}{2} \left[\begin{array}{ccc} \frac{1}{3} & - & \frac{1}{5} \\ & + & \frac{1}{5} & - & \frac{1}{7} \\ & & \vdots & & \\ & & + \frac{1}{2N-1} & - & \frac{1}{2N+1} \\ & & + \frac{1}{2N+1} & - & \frac{1}{2N+3} \end{array} \right]$ $= \frac{1}{2} \left[\frac{1}{3} - \frac{1}{2N+3} \right] = \frac{1}{6} - \frac{1}{2(2N+3)}$	
(iv)	$\sum_{n=1}^N \frac{1}{4n^2 - 1} = \sum_{t=0}^{N-1} \frac{1}{4(t+1)^2 - 1}$ $= \frac{1}{3} + \sum_{t=1}^{N-1} \frac{1}{4(t+1)^2 - 1}$ $= \frac{1}{3} + \frac{1}{6} - \frac{1}{2(2(N-1)+3)}$ $= \frac{1}{2} - \frac{1}{2(2N+1)}$	
13(i)	$\frac{x}{2} = -1 - z, y = 0 \Rightarrow \vec{r} = \begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix}$ Vector parallel to $\pi_2 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$ $\vec{n} = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \times \begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}$ $\pi_1 : \vec{r} \bullet \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix} \bullet \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} \Rightarrow \vec{r} \bullet \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} = -2$	
(ii)	Let foot of perpendicular from Q to π_2 be N.	

	$\overrightarrow{ON} = \begin{pmatrix} 3 \\ 13 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}$ $\begin{pmatrix} 3+\lambda \\ 13+3\lambda \\ 6+2\lambda \end{pmatrix} \bullet \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} = 2$ $\Rightarrow \lambda = -4$ $\overrightarrow{ON} = \begin{pmatrix} -1 \\ 1 \\ -2 \end{pmatrix}$	
(iii)	$\overrightarrow{ON} = \frac{\overrightarrow{OQ} + \overrightarrow{OQ'}}{2} \Rightarrow \overrightarrow{OQ'} = 2 \begin{pmatrix} -1 \\ 1 \\ -2 \end{pmatrix} - \begin{pmatrix} 3 \\ 13 \\ 6 \end{pmatrix} = \begin{pmatrix} -5 \\ -11 \\ -10 \end{pmatrix}$ $\overrightarrow{PQ'} = \begin{pmatrix} -5 \\ -11 \\ -10 \end{pmatrix} - \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} = \begin{pmatrix} -6 \\ -10 \\ -10 \end{pmatrix} = -2 \begin{pmatrix} 3 \\ 5 \\ 5 \end{pmatrix}$ $\therefore l_2 : \vec{r} = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 5 \\ 5 \end{pmatrix} \quad \text{OR} \quad \therefore l_2 : \vec{r} = \begin{pmatrix} -5 \\ -11 \\ -10 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 5 \\ 5 \end{pmatrix}$	
(iv)	$\begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} = k \begin{pmatrix} a \\ 6 \\ 4 \end{pmatrix} \Rightarrow a = 2$ Method 1: $\pi_1 : r \bullet \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} = -2 \quad \pi_2 : r \bullet \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} = \frac{b}{2}$ $\text{Distance between the 2 planes} = \left \frac{\frac{b}{2} - (-2)}{\sqrt{1+9+4}} \right = \sqrt{224} \Rightarrow b = 108 \text{ or } -116$ Method 2: $\text{Distance QN} = \sqrt{(3-(-1))^2 + (13-1)^2 + (6-(-2))^2} = \sqrt{224}$ $b = \overrightarrow{OQ} \bullet \begin{pmatrix} 2 \\ 6 \\ 4 \end{pmatrix} = \begin{pmatrix} 3 \\ 13 \\ 6 \end{pmatrix} \bullet \begin{pmatrix} 2 \\ 6 \\ 4 \end{pmatrix} = 108 \quad \text{OR} \quad b = \overrightarrow{OQ'} \bullet \begin{pmatrix} 2 \\ 6 \\ 4 \end{pmatrix} = \begin{pmatrix} -5 \\ -11 \\ -10 \end{pmatrix} \bullet \begin{pmatrix} 2 \\ 6 \\ 4 \end{pmatrix} = -116$ Method 3 $\pi_1 : r \bullet \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} = -2 \Rightarrow r \bullet \frac{1}{\sqrt{14}} \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} = -\frac{2}{\sqrt{14}}$	

	$\pi_2 : r \bullet \frac{1}{\sqrt{14}} \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} = -\frac{2}{\sqrt{14}} \pm \sqrt{224}$ $\Rightarrow r \bullet \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} = -2 \pm 56 = 54 \quad \text{or} \quad -58 \Rightarrow b = 108 \quad \text{or} \quad -116$	
(v)	$a \neq 2, b \in \Re$	
14 (a)(i)	Let $3(y-1) = 5\sin\theta \Rightarrow \frac{dy}{d\theta} = \frac{5}{3}\cos\theta$ When $y = \frac{1}{6}$, $\sin\theta = -\frac{1}{2} \Rightarrow \theta = -\frac{\pi}{6}$. When $y = \frac{8}{3}$, $\sin\theta = 1 \Rightarrow \theta = \frac{\pi}{2}$. $\begin{aligned} \therefore \int_{\frac{1}{6}}^{\frac{8}{3}} \sqrt{25-9(y-1)^2} dy &= \int_{-\frac{\pi}{6}}^{\frac{\pi}{2}} \sqrt{25-25(\sin\theta)^2} \times \frac{5}{3} \cos\theta d\theta \\ &= \frac{25}{3} \int_{-\frac{\pi}{6}}^{\frac{\pi}{2}} \frac{1+\cos 2\theta}{2} d\theta \\ &= \frac{25}{6} \left[\theta + \frac{1}{2} \sin 2\theta \right]_{-\frac{\pi}{6}}^{\frac{\pi}{2}} \\ &= \frac{25}{6} \left[\left(\frac{\pi}{2} + \frac{1}{2} \sin \pi \right) - \left(-\frac{\pi}{6} - \frac{1}{2} \sin \frac{\pi}{3} \right) \right] \\ &= \frac{25}{6} \left(\frac{2\pi}{3} + \frac{\sqrt{3}}{4} \right) = 25 \left(\frac{\pi}{9} + \frac{\sqrt{3}}{24} \right) \end{aligned}$	
(a) (ii).	$\begin{aligned} \text{Area} &= \int_{\frac{1}{6}}^{\frac{8}{3}} \left[\frac{3y-6}{2} - \left(1 - \sqrt{25-9(y-1)^2} \right) \right] dy \\ &= \int_{\frac{1}{6}}^{\frac{8}{3}} \left(\frac{3y-8}{2} \right) dy + \int_{\frac{1}{6}}^{\frac{8}{3}} \sqrt{25-9(y-1)^2} dy \\ &= \frac{1}{2} \left[\frac{3y^2}{2} - 8y \right]_{\frac{1}{6}}^{\frac{8}{3}} + 25 \left(\frac{\pi}{9} + \frac{\sqrt{3}}{24} \right) \\ &= -\frac{75}{16} + 25 \left(\frac{\pi}{9} + \frac{\sqrt{3}}{24} \right) \\ &= 25 \left(\frac{\pi}{9} + \frac{\sqrt{3}}{24} - \frac{3}{16} \right) \text{ square units} \end{aligned}$	
(b)	$V_1 = \pi \int_{-4}^1 \left[1 + \frac{\sqrt{25-(x-1)^2}}{3} \right]^2 dx = 85.9201$ $V_2 = \pi \int_{-4}^{-3} \left[1 - \frac{\sqrt{25-(x-1)^2}}{3} \right]^2 dx = 0.490117 \quad \text{B1}$ $V_3 = \pi \int_{-3}^1 \left[\frac{2x+6}{3} \right]^2 dx = 29.78695$ Volume required = $V_1 - V_2 - V_3 = 55.6$.	